

# Empirical Industrial Organization

CY Cergy Paris Université  
Master in economic analysis and PhD, Economics track.  
Problem Set 2  
Due: February 4, 2026

## Problem 1

A seller auctions off an object to 3 buyers. Buyer valuations  $\epsilon_\ell, \ell = 1, 2, 3$ , are distributed i.i.d. with support  $[0, 1]$  and distribution function  $F(x) = x^2$ . Each buyer  $\ell$  knows the realization of  $\epsilon_\ell$  but only knows the distribution of other buyers' valuations.

1. Determine the distribution function of the maximum valuation among 2 buyers.

The item is sold through a first price sealed bid auction. We look for a symmetric Bayes-Nash equilibrium where all bidders use the same bidding function  $\beta$ , where a bidder with value  $\epsilon$  bids  $b = \beta(\epsilon)$ . The function  $\beta$  is taken to be strictly increasing and differentiable.

2. Write bidder 1's problem and use the corresponding first order condition to determine the equilibrium bidding function  $\beta$  (Instruction: you should write all the equations using the distribution function you derived in question 1. Do not use the general formulation that is in the slides.)
3. Assume now that a sealed bid second price auction with reserve price  $r$  is used instead. Show that the optimal reserve price is  $r = \frac{1}{\sqrt{3}}$ .